

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCC3073 DATA SCIENCE

THURSDAY, 22 SEPTEMBER 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the **FIRST (1)** question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

UCCC3073 DATA SCIENCE**SECTION A (Answer ALL questions)**

- Q1. (a) With the growth of digitalized business models, more data are collected to maximise the gain from the business models.

Consider McDonald's fast-food restaurant:

- (i) Identify and explain any TWO (2) characteristics of big data that are important for McDonald's. In your answer, provide an example for each of the identified characteristics in the context of McDonald's. (6 marks)
- (ii) Provide any TWO (2) examples of data that should be collected by McDonald's to perform big data analysis. (2 marks)
- (iii) Describe any TWO (2) benefits that can be gained by McDonald's for having such big data analysis. (2 marks)

- (b) Table 1.1 shows a data containing 4 samples ($p1, p2, p3, p4$) and 2 attributes (x, y).

Table 1.1

	x	y
$p1$	2	2.2
$p2$	1.6	1.8
$p3$	1.9	2
$p4$	1.5	1

Given a new sample, $p5 = (1.1, 1.4)$ as a query, rank the samples in the data set based on the similarity with the query using the following methods. Round your answer to the nearest 2 decimal points and show all the working steps.

- (i) Euclidean distance. (6 marks)
- (ii) Manhattan distance. (2 marks)
- (iii) Supremum distance. (2 marks)

UCCC3073 DATA SCIENCE**Q1. (Continued)**

- (c) A survey was conducted by the Department of Statistics Malaysia to find out the Malaysians response on the lift of travel ban on 1 April 2022. 1,954 adults responded to the survey. A descriptive summary on the frequency, relative frequency and percent frequency distributions of each question asked in the survey were computed. A representative portion of the analysis is given in Figure 1.1.

Survey Question 1: Do you agree for government to lift travel's restriction?

Lift of travel restriction	Frequency	Relative	Percentage
Strongly agree	840	0.430	43.0
Agree	560	0.287	28.7
Disagree	470	0.240	24.0
Strongly disagree	84	0.043	4.3

Survey Question 2: Do you agree that lift of travel restriction will improve local economy?

Improve economy	Frequency	Relative	Percentage
Strongly agree	1,500	0.768	76.8
Agree	319	0.163	16.3
Disagree	125	0.064	6.4
Strongly disagree	10	0.005	0.5

Survey Question 3: Do you agree that lift of travel restriction will increase the COVID cases?

Improve economy	Frequency	Relative	Percentage
Strongly agree	250	0.128	12.8
Agree	715	0.366	36.6
Disagree	524	0.268	26.8
Strongly disagree	465	0.238	23.8

Figure 1.1

- (i) Identify sample and population of the survey. (1 mark)
- (ii) Derive THREE (3) insights from Figure 1.1. (3 marks)
- (iii) Suggest ONE (1) appropriate type of chart to present all the information shown in Figure 1.1. Justify your answer. (1 mark)
- [Total : 25 marks]

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- Q2. Table 2.1 shows the data on the amount of time an engineer required to repair a photocopy machine.

Table 2.1

Number of parts to be repaired (x)	1	2	3	4	5
Minutes taken	13	22	40	59	86

- (a) Regression analysis is considered to be the best approach for analysing this data. Explain why regression is appropriate for this data. (2 marks)
- (b) Determine the best-fit regression line for the data. Show all the working steps in your answer. (8 marks)
- (c) Apply the regression line in Q2(b) to determine the amount of time needed (in minutes) for the engineer to repair 9 broken parts in the photocopy machine. Comment on the reliability of your prediction. (5 marks)
- (d) Determine the coefficient of determination, R^2 , for the predicted regression line in Q2(b) and use it to comment on the reliability of your prediction in Q2(c). Round your answer to the nearest 3 decimal points and show all the working steps. (8 marks)
- (e) Suggest any TWO (2) ways to evaluate model fitness for regression model. (2 marks)

[Total : 25 marks]

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- Q3. Consider the following transactional data from a computer shop where 1, 2, 3, 4, 5, 6, 7 are items sold in the shop.

Table 3.1

Transaction ID	Items
T1	1, 2, 3
T2	1, 3, 5, 6, 7
T1	1, 2, 3, 7
T4	2, 5
T5	4, 5, 7

Assume that the minimum support is 40% and confidence is 70%

- (a) Find all the frequent item sets. In your answer, show the candidate item sets obtained at each step of the pruning stage and list down the all the frequent item sets. (10 marks)
- (b) Generate the Apriori rules for frequent item set L3. In your answer, show the working steps on how you obtain the rules and list down the finalised rules. (8 marks)
- (c) Describe TWO (2) interesting trends that you observe from the Apriori rules that you have generated in Q3(b). (4 marks)
- (d) Based on your observations on Q3(c), provide ONE (1) suggestion to the retailer to improve the sales. (3 marks)

[Total : 25 marks]

UCCC3073 DATA SCIENCE**SECTION B (Choose only ONE question)**

- Q4. Table 4.1 shows the confusion table for the naïve bayes (NB) models with and without misclassification cost applied. Model 1 is the NB with equal cost. Model 2 is the NB with unequal cost of which the cost for false positive ($Cost_{FP}$) is 20 and the cost for true positive ($Cost_{TP}$) is -10.

Table 4.1

N = 1,000		Prediction values			
		Model 1		Model 2	
		No	Yes	No	Yes
Actual values	No	1000	0	900	100
	Yes	7000	2000	1800	7200

- (a) Calculate the baseline performance. (2 marks)
- (b) Calculate the following evaluation measures for both models. For each measure, identify the best model. Round your answer to the nearest 4 decimal points.
- (i) Misclassification rate. (3 marks)
- (ii) Precision. (3 marks)
- (iii) F_1 score. (3 marks)
- (iv) Profit cost per record. (3 marks)
- (c) Using the computed results in Q4(a) and Q4(b), justify whether you would propose the model to the company. If yes, identify which model you would propose. (5 marks)
- (d) Suggest TWO (2) ways to overcome the issues you have observed in Q4(c). (6 marks)

[Total : 25 marks]

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- Q5. You are required to propose a diagnostic model for a hospital to screen cancer patients. Due to a lack of book-keeping system in the hospital, you are given only 60 patient samples to build the diagnostic model. You decided to use 50 samples for training the model and the remaining 10 samples for testing the model. In the model development, you constructed 2 models, namely Naïve Bayes (NB) and K nearest neighbour (KNN). Table 5.1 is the prediction results of the testing set made by NB and KNN.

Table 5.1

Patient Id	Actual diagnosis	NB Prediction	KNN Prediction
P1	Healthy	Cancer	Cancer
P2	Healthy	Healthy	Cancer
P3	Healthy	Healthy	Cancer
P4	Healthy	Healthy	Healthy
P5	Cancer	Cancer	Cancer
P6	Cancer	Cancer	Cancer
P7	Cancer	Cancer	Healthy
P8	Cancer	Cancer	Healthy
P9	Cancer	Healthy	Healthy
P10	Cancer	Cancer	Healthy

- (a) Compute the confusion matrix for both the NB and KNN models. (4 marks)
 - (b) Calculate the baseline performance. (2 marks)
 - (c) Calculate the following evaluation measures for both models. For each measure, identify the best model. Round your answer to the nearest 4 decimal points.
 - (i) Classification accuracy. (3 marks)
 - (ii) Precision. (3 marks)
 - (iii) Recall. (3 marks)
 - (d) Using the computed results in Q5(b) and Q5(c), justify which model is a better fit for the data. (5 marks)
 - (e) Justify whether you would recommend the model identified in Q5(d) to the hospital. (3 marks)
 - (f) Suggest TWO (2) alternate evaluation metrics to consolidate your findings in Q5(c). (2 marks)
- [Total : 25 marks]

APPENDIX

Regression	
Straight line, $\hat{y} = \beta_0 + \beta_1 x$	t value, $t = \frac{\hat{\beta}}{SE}$
Coefficients $\hat{\beta}_1 = \frac{p \sum xy - \sum x \sum y}{p \sum x^2 - (\sum x)^2}$	Standard errors of coefficients $SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$
$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$,	$SE(\hat{\beta}_0) = s \sqrt{\frac{1}{p} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$
Coefficient of Determination $R^2 = \left[\frac{p \sum xy - \sum x \sum y}{\sqrt{p \sum x^2 - (\sum x)^2} * \sqrt{p \sum y^2 - (\sum y)^2}} \right]^2$	Residuals $\hat{\epsilon} = y - \hat{y}$
$R = \sqrt{R^2}$	$s^2 = \frac{\sum \hat{\epsilon}^2}{p - 2}$
	$s = \sqrt{s^2}$

Naïve Bayes
Bayes' rule, $P(H E) = \frac{P(E H)P(H)}{P(E)}$
Laplace correction, $\frac{x_i+1}{c+\alpha}$
Probability density function for a Gaussian distribution, $pdf(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\bar{x})^2}{2\sigma^2}}$

Association Rules
Support, $s = \frac{freq(X,Y)}{N}$
Confidence, $c = \frac{freq(X,Y)}{freq(X)}$
Lift, $l = \frac{Support}{Supp(X)*Supp(Y)}$

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Evaluation metrics	
Classification accuracy, $acc = \frac{TN+TP}{TN+TP+FN+FP}$	Precision, $PPV = \frac{TP}{TP+FP} = 1 - FDR$
Misclassification rate, $1 - acc$	F1 score, $F_1 = \frac{2TP}{2TP+FP+FN}$
Sensitivity / Recall, $recall = \frac{TP}{TP+FN}$	Overall model cost $= TN.Cost_{TN} + FP.Cost_{FP} + FN.Cost_{FN} + TP.Cost_{TP}$
Specificity, $spec = \frac{TN}{TN+FP}$	Model cost per case $= \frac{Overall\ model\ cost}{N}$
False discovery rate, $FDR = \frac{FP}{FP+TP}$	Model profit per case $= -model\ cost\ per\ case$

Proximity measures	
Euclidean distance, $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$	Hamming distance, $d(a, b) = (a \oplus b)$
Manhattan distance, $d(x, y) = \sum_{i=1}^n x_i - y_i $ $d(x, y) = \sum_{i=1}^n x_i - x + y_i - y $	Simple matching coefficient, $SMC = \frac{num\ of\ matching\ attribute\ values}{num\ of\ attributes}$
Supremum distance, $d(x, y) = \max(x_i - y_i)$ $d(x, y) = \max(x_i - x , y_i - y)$	Jaccard coefficient, $J = \frac{num\ of\ matching\ presences}{num\ of\ attributes\ not\ involved\ in\ 00\ matches}$
	Cosine similarity, $cos(x, y) = \frac{x \cdot y}{ x y }$

Miscellaneous	
Min-Max normalisation, $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$	Confidence intervals, $CI = \bar{x} \pm z * SE$
z-score normalisation, $x' = \frac{x - \mu}{\sigma}$	Standard error, $SE = \frac{\sigma}{\sqrt{n}}$
Sample mean, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Z-test, $Z = \frac{\bar{x} - \mu}{SE_{\bar{x}}}$
Sample standard deviation, $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$	

